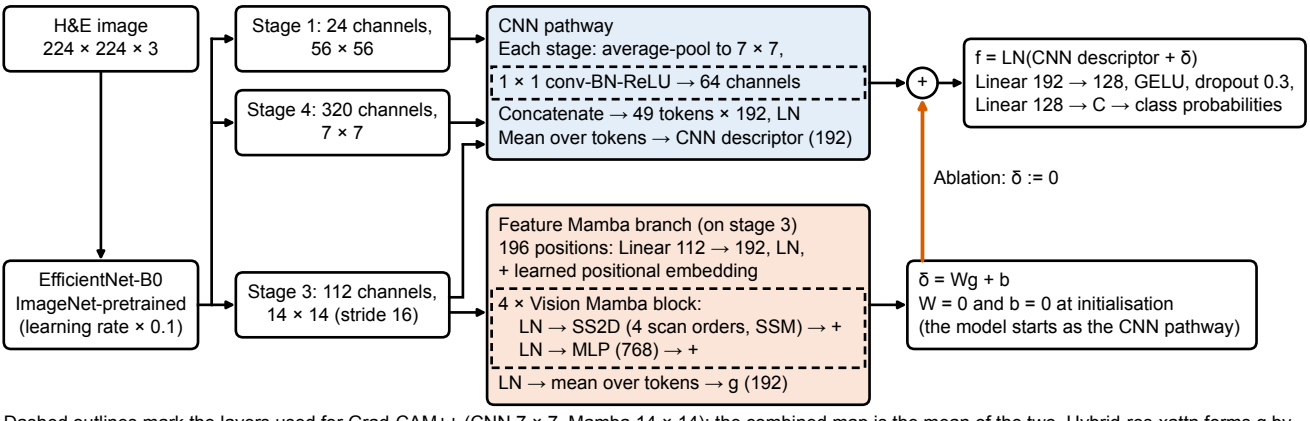
a Residual CNN–Mamba hybrid (hybrid-res)



Dashed outlines mark the layers used for Grad-CAM++ (CNN 7 × 7, Mamba 14 × 14); the combined map is the mean of the two. Hybrid-res-xattn forms g by gated bidirectional cross-attention between the 49 CNN and 196 Mamba tokens (4 heads); the variant was chosen on validation accuracy only.

b Evaluation design (one training recipe for every model)

LC25000 (15,000 images, three classes)
- Source groups recovered by an embedding shortlist and SIFT/RANSAC: 716 groups (ARI 0.949 with LC25000-clean)
- Grouped 70/15/15 split, three seeds (leakage-controlled)
- Per-file split (leaky reference)
- Inference unit: source group (about 107 per test set)
- Clean-subset sensitivity analysis

LungHist700 (691 images, 45 patients)
- Three classes: patient-level five-fold CV; image-level five-fold CV as the leaky reference
- Seven-class differentiation grading: patient-level five-fold CV
- Pooled out-of-fold predictions; inference unit: patient
- 20× and 40× together; a patient never spans folds

Transfer, explanations and cost
- External test: LC25000-trained models on LungHist700 at 20× and 40×, without adaptation
- Pathway attribution: size and removal of δ
- Faithfulness: deletion and insertion AUC against a random map, with two deletion canvases
- Comparators: CNN pathway, original hybrid, EfficientNet-B0, ViT-S, Swin-T, VMamba-T, Phikon probe
- Parameters, operations, latency and memory